A lower bound on the mean value of the Erdős-Hooley delta function

Terence Tao · 23 August 2023

https://terrytao.wordpress.com/2023/08/23/a-lower-bound-on-the-mean-value-of-the-erdos-hooley-delta-function/

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Kevin Ford, Dimitris Koukoulopoulos and I have just uploaded to the arXiv our paper "A lower bound on the mean value of the Erdős-Hooley delta function". This paper complements the recent paper of Dimitris and myself obtaining the upper bound

$$\frac{1}{x}\sum_{n\leq x}\Delta(n) \ll (\log\log x)^{11/4}$$

on the mean value of the *Erdős-Hooley delta function*

$$\Delta(n) := \sup_u \#\{d \mid n : e^u < d \leq e^{u+1}\}$$

In this paper we obtain a lower bound

$$\frac{1}{x}\sum_{n\leq x}\Delta(n) \gg (\log\log x)^{1+\eta-o(1)}$$

where $\eta = 0.3533227\ldots$ is an exponent that arose in previous work of result of Ford, Green, and Koukoulopoulos, who showed that

$$\Delta(n) \gg (\log\log n)^{\eta-o(1)} \tag{1}$$

for all n outside of a set of density zero. The previous best known lower bound for the mean value was

$$\frac{1}{x}\sum_{n\leq x}\Delta(n) \gg \log\log x,$$

due to Hall and Tenenbaum.

The point is the main contributions to the mean value of $\Delta(n)$ are driven not by "typical" numbers n of some size x, but rather of numbers that have a splitting

$$n = n'n''$$

where n'' is the product of primes between some intermediate threshold $1 \leq y \leq x$ and x and behaves "typically" (so in particular, it has about $\log\log x - \log\log y + O(\sqrt{\log\log x})$ prime factors, as per the Hardy-Ramanujan law and the Erdős-Kac law, but n' is the product of primes up to y and has double the number of typical prime factors – $2\log\log y + O(\sqrt{\log\log x})$, rather than $\log\log y + O(\sqrt{\log\log x})$ – thus n'' is the type of number that would make a significant contribution to the mean value of the divisor function $\tau(n'')$. Here y is such that $\log\log y$ is an integer in the range

$$\varepsilon\log\log x \leq \log\log y \leq (1-\varepsilon)\log\log x$$

for some small constant $\varepsilon > 0$ there are basically $\log\log x$ different values of y give essentially disjoint contributions. From the easy inequalities

$$\Delta(n) \gg \Delta(n')\Delta(n'') \geq \frac{\tau(n')}{\log n'}\Delta(n'') \tag{2}$$

(the latter coming from the pigeonhole principle) and the fact that $\frac{\tau(n')}{\log n'}$ has mean about one, one would expect to get the above result provided that one could get a lower bound of the form

$$\Delta(n'') \gg (\log\log n'')^{\eta - o(1)} \tag{3}$$

for most typical n'' with prime factors between y and x. Unfortunately, due to the lack of small prime factors in n'', the arguments of Ford, Green, Koukoulopoulos that give (1) for typical n do not quite work for the rougher numbers n''. However, it turns out that one can get around this problem by replacing (2) by the more efficient inequality

$$\Delta(n) \gg \frac{\tau(n')}{\log n'} \Delta^{(\log n')}(n'')$$

where

$$\Delta^{(v)}(n) := \sup_u \#\{d \mid n : e^u < d \leq e^{u+v}\}$$

is an enlarged version of $\Delta^{(n)}$ when $v \geq 1$. This inequality is easily proven by applying the pigeonhole principle to the factors of n of the form $d'd''$, where d' is one of the $\tau(n')$ factors of n', and d'' is one of the $\Delta^{(\log n')}(n'')$ factors of n'' in the optimal interval $[e^u, e^{u+\log n'}]$. The extra room provided by the enlargement of the range $[e^u, e^{u+1}]$ to $[e^u, e^{u+\log n'}]$ turns out to be sufficient to adapt the Ford-Green-Koukoulopoulos argument to the rough setting. In fact we are able to use the main technical estimate from that paper as a "black box", namely that if one considers a random subset A of $[D^c, D]$ for some small $c > 0$ and sufficiently large D with each $n \in [D^c, D]$ lying in A with an independent probability $1/n$, then with high probability there should be $\gg c^{-1/\eta + o(1)}$ subset sums of A that attain the same value. (Initially, what "high probability" means is just "close to 1", but one can reduce the failure probability significantly as $c \to 0$ by a "tensor power trick" taking advantage of Bennett's inequality.)

Comments (9)

isaacg227 · 23 August 2023

Looks like there's a typo in the equation just after "In this paper we obtain a lower bound ". Thanks for the post!
[Corrected, thanks – T.]

isaacg227 · 23 August 2023

Similarly, the 1/x in front of the summations in the first two inequalities seem to be redundant with the x in front of the log log ... on the right hand side of those inequalities.
[Corrected, thanks – T.]

joe34 · 23 August 2023

Is there a conjecture for tight bounds, or even an exact growth rate, between the 11/4=3.75 and the 1.353...? And what is your hunch on the sort of difficulty it requires? Thank you!

> **Terence Tao** · 24 August 2023
>
> Good question. My co-authors, who have worked more extensively with the Erdős-Hooley function than I have, are more inclined to conjecture that the exponent 1.353... is in fact the truth. I do not have a strong opinion here, though certainly I do not think the 11/4 exponent is optimal, as it comes from some clear limitations to our upper bound method (actually, I am surprised that our upper bound approach, based on the moment method, is as effective as it is).

Anonymous · 24 August 2023

It would be interesting to consider this problem for a generalization of the function $\Delta(n)$ to $\Delta(n,\delta)$ where in the definition of $\Delta(n)$ the upper bound e^{u+1} is replaced by $e^{u+\delta}$ for some fixed positive parameter δ (i.e. using a general "window" for the divisors d inthe definition)

> **Terence Tao** · 24 August 2023
>
> For $\delta \geq 1$, one has from the triangle inequality that $\Delta(n) \leq \Delta(n,\delta) \ll \delta\Delta(n)$, while similarly for $0 < \delta \leq 1$ one has $\delta\Delta(n) \ll \Delta(n,\delta) \leq \Delta(n)$. So for fixed δ the asymptotics are not much different from the $\delta = 1$ case, but it could be interesting to consider values of δ that grow or shrink with n. For instance in the arguments of our paper it becomes important to consider a choice of δ which is comparable to $\log n$.

Anonymous · 24 August 2023

There appears to be a couple of typos with the co-author's surname. It should be Koukoulopoulos instead of Koukoulopolous.
[Corrected, thanks – T.]

Zach Hunter · 24 August 2023

Small typo in your writeup: "area" is written rather than "are" in Definition 1 on page 2.
[Thanks, this will be corrected in the next revision of the ms. -T]

ducduc2710 · 8 November 2023

Can we use the ratio σ(n+1)/σ(n) and the Lagarias inequality to prove RH? Dear Professor Terence Tao